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"From our perspective, quality standards within the India automotive industries are at par with any emerging developing country standards such as BRIC countries," opines Kumar.

Sustainable mobility

India is now paving the way forward in future mobility. Technological advancements are changing the way we think about cars in terms of energy efficiency, safety, comfort, car-to-x communication and even infotainment solutions, explains Ramachandra.

In future, security and connectivity would need to run hand in hand. For example, a user entering the car and connecting his personal device to the head unit would compromise vehicle security. While improved connectivity enables the car to access information continuously from the cloud and provide enhanced services to the driver and passengers, it also becomes vulnerable to potential hacking and even tampering of the controls of the vehicle, endangering its occupants. Pai suggests, enhanced security platforms such as secured gateway modules could act as a central hub and a gatekeeper between the vehicle and the outside world to prevent unauthorised access.

The ever-increasing functionality, applications, convenience and user-friendliness of smartphones and tablets have influenced the automotive industry to work on integrating these devices or similar functionality into the vehicle.

Presently, these devices can be paired to the head unit of the car and engage in a whole set of functions like voice calls, navigation and audio/video content. "Connected cars of the future will seamlessly integrate smartphones, wearables and sensors into the car, and provide users not only with applications such as cloud based navigation, real-time traffic updates, fuel reminders, location of parked vehicles and emergency/service assist, but also provide features such as active health monitoring

Facts and figures: Indian automotive electronics market

- Automotive electronics account for around five per cent of the total semiconductor market size of US\$ 7 billion
- On average, total electronics content of an Indian vehicle is around ten per cent, with an average value of US\$ 50 per car and US\$ 5 per two-wheeler

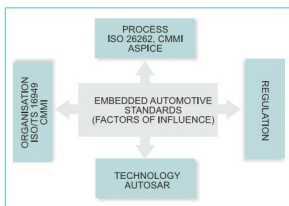


Fig. 3: Automotive electronics standards (Source: NASCOM)

to systems like lane-departure warning system that alerts the driver if he/she veers away from lanes, drowsiness detection system, blind-spot monitoring system that enables the driver to view areas around the car that are not viewable with standard mirrors and so on. Another area that might see additional inclusion of semiconductors will be in body control modules (BCMs), integrated systems for controlling various body peripherals like lighting, wipers and HVACs.

Harmonised development through standards

The growing amount of electronics in power train, chassis, body control

and infotainment applications in passenger cars and vehicles in India make it increasingly challenging to manage development of complex embedded systems that integrate software and hardware from a large set of suppliers. Therefore adoption of globally-established standards will enable the industry to harmonise their development activities to compete effectively.

Standards governing the automotive electronics landscape can be presented as a triad of technology, process and organisation (Fig. 3). It is acknowledged that, standards are also driven by government policy and regulations (for example, legislative and emission), keeping in mind future demands.

Both Merli and Ramachandra point out that the significance of standards must be taken into consideration if the Indian automotive industry has to cope with the increasing vehicle system complexity and an increase in safety-relevant functions in embedded systems.